

Sparse Demonstrations

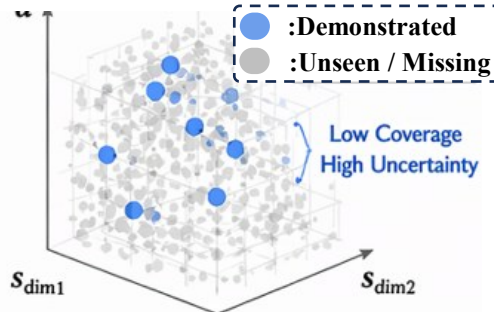
Only a few trajectories lead state-action space is severely under-covered

Front

Wrist

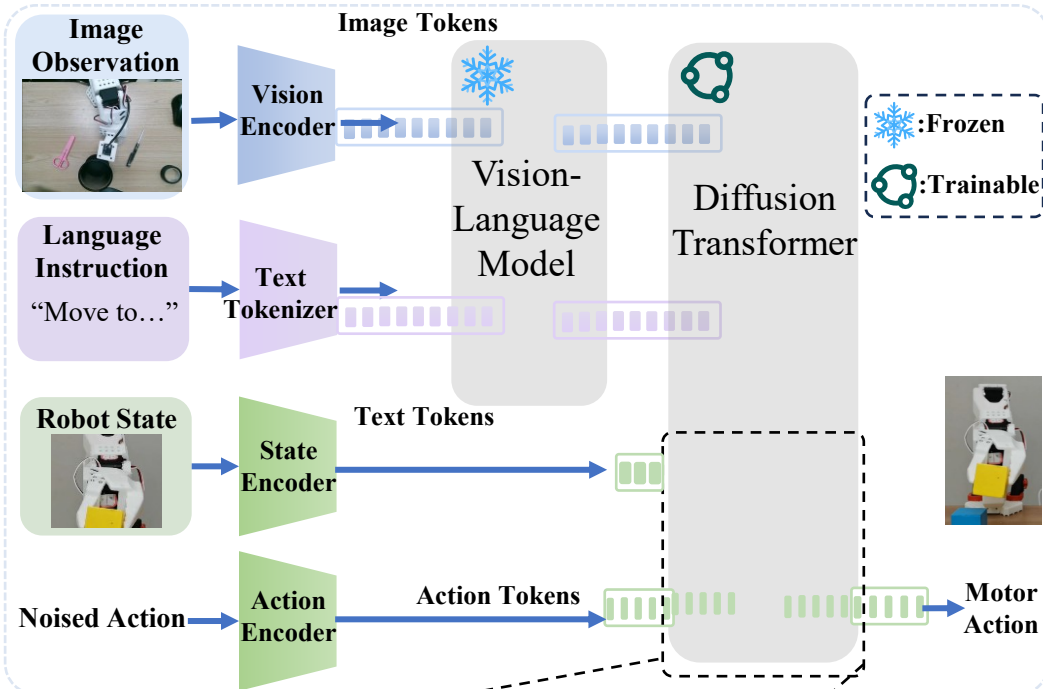


State-Action Space(s, a)



Latent Policy Modeling

VLA policy learns a mapping from multi-modal inputs to a distribution over actions, enabling robust learning from sparse data.

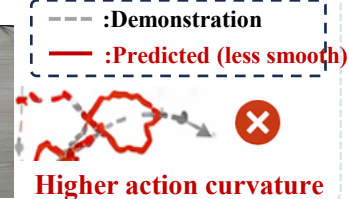
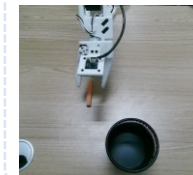


Training Regularization

Both variants share the same architecture and inference path; only the training objective differs.

Without Training Regularization

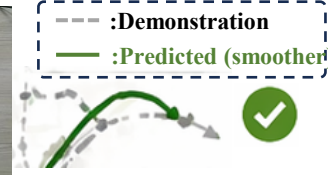
Base flow-matching loss



Higher action curvature

With Training Regularization

Base + temporal/velocity terms



Lower action curvature

Robot State q_t

State Encoder



Noised Action

at

at+1

...

at+H-1

Action Encoder



Cross-Attention

Self-Attention

Cross-Attention

Self-Attention

Output Projection

$\times N$ (DiT Blocks)

K iterations



Action Decoder

Inference

at

at+1

...

at+H-1



Motor Action

Training Only

Training-only Objective

$$\mathcal{L}_{total} = \mathcal{L}_{flow} + \lambda_t \mathcal{L}_{temp} + \lambda_v \mathcal{L}_{vel}$$